

HYDRAULIC FRACTURING PROPOSAL-V1



CLIENT : AGIBA Petroleum Company

WELL : **RAML 17**

ZONE : Bahareya III

PREPARED FOR : Agiba Stimulation Team

PREPARED BY : AHMED SAMIR ZAKI < AZaki2@slb.com , Mobile: 0170001372>

SUBMITTED DATE : **2ND DECEMBER 2011**

2ND NOVEMBER 2011

Dear Sirs,

Schlumberger would like to thank you for the opportunity to design a stimulation treatment for well RAML17 in Raml field. In the following report, you will find the preliminary fracturing Design and program.

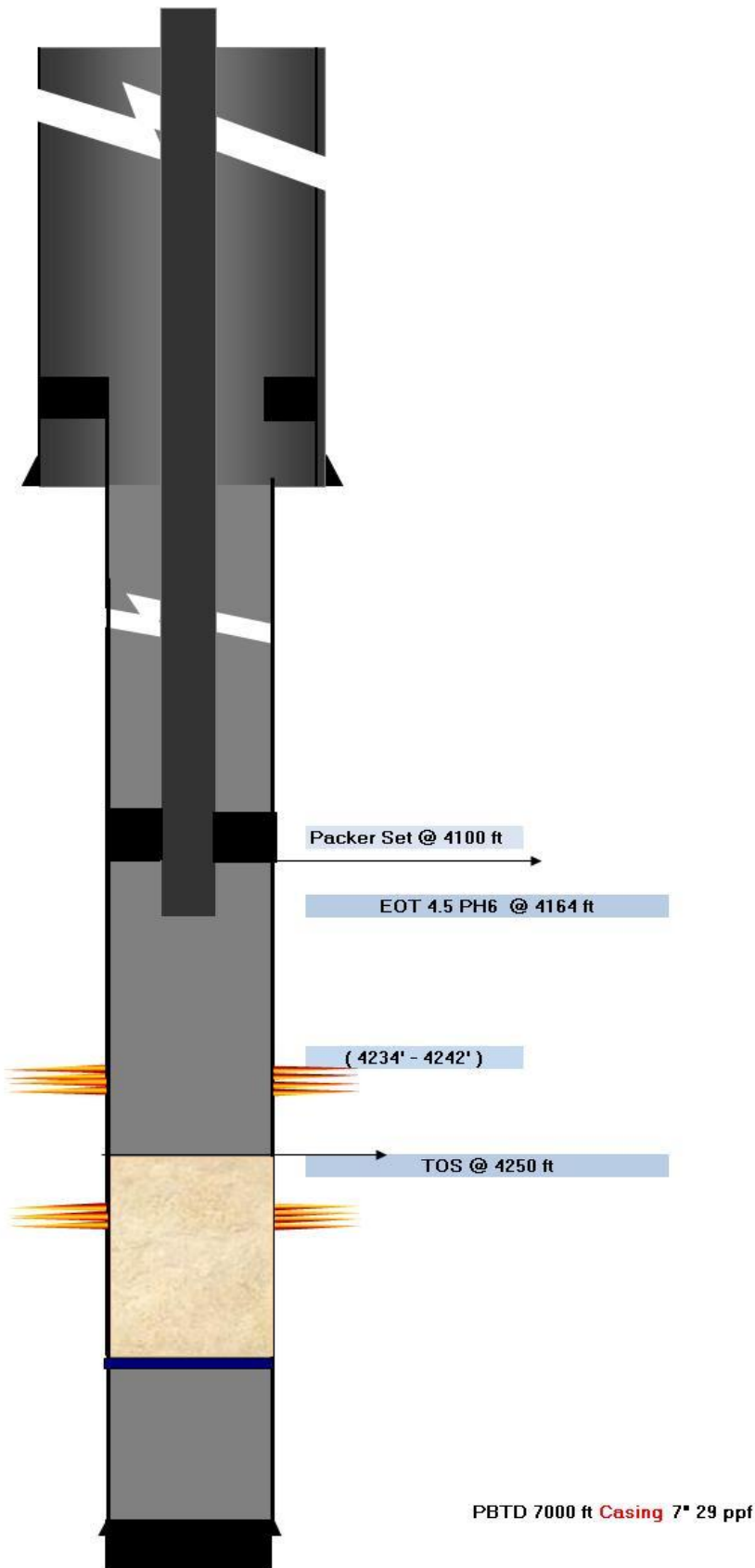
Schlumberger is offering a hydraulic proppant fracturing treatment-using ClearFRAC XT along with 20/40 S580 proppant to maximize conductivity.

Please do not hesitate to contact us if you have any inquiry regarding the content of this report or if you need further information.

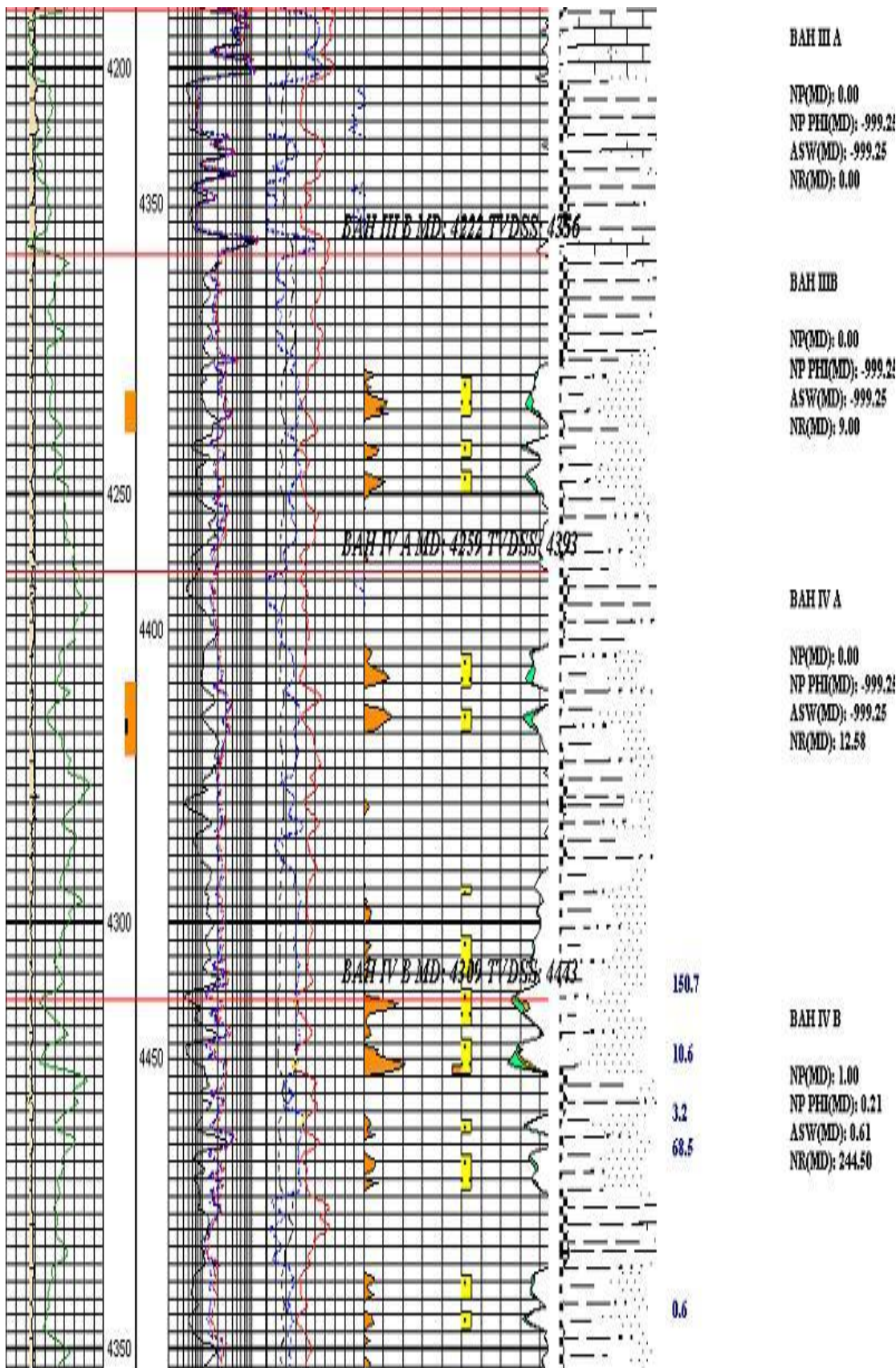
Yours sincerely,

Ahmed Samir Zaki
Stimulation Technical Engineer
Schlumberger

WELL SKETCH:



WELL LOG:



GENERAL PROCEDURES

Running the Frac String:

- ▣ Pressure Test the Frac String to **8,000 psi** against Ceramic Disk
- ▣ Frac string is 4.5 PH6
- ▣ Connection required at surface is 4.5 PH6

Fracturing Operations:

1. Perform SLB pre-rig up safety meeting.
2. **Perform Sand Plug** by Frac equipments as calculated below, However Sand Plug will be recalculated based on actual data on location with AGIBA job witness:

Casing OD (inch)		7
Casing Weight		29
Casing ID (inch)		6.184
Casing Capacity (bbl/ft)		0.03714966
TOS (ft)	4250	ft
PBTD (ft)	4297	ft
PPA		2
Rate (bpm)		3.5
Top Perf	4272	ft
Bottom Perf	4280	ft

Bulk Density (lb/gal)		15.012
Apparent Density (lb/gal)		26.9382
Height of SP (ft)		47
Sand Plug Vol. (gal)		73.33342116
Slurry Yield		1.07424401
CFR		0.930887201

Mass of Prop. Sand Plug (lb)		1100.881
Clean fluid needed @ SP stg.(gal)		550.4407
Slurry needed @ SP stg.(gal)		591.3076

3. R/U all fracturing surface equipment. (Frac tanks, frac pumps, etc...)
 - Tanks should have been mixed with 4 % KCL water upon filling the tanks
 - Load Proppant into Sand chief as specified in the design.
4. Batch mix gel in the Frac Tanks and X-linker solution in the RMX.
5. Measure fracture fluid properties and compare to published data to confirm fluid is within specification: Base fluid viscosity, temperature and pH, Crosslinking, delay time and pH

6. Hold Safety meeting with all key personnel on location and assign the job responsibility
7. Flush all lines with water

8. Pressure Test of Main Line:

- Close Dowell Gate valve on the rig floor. Pressures test the surface lines to 9000 psi against Gate valve for 10 minutes. Visually inspect surface lines for leaks. Bleed off pressure.
- Pumps Automatic Shut Down should be set @ 8000 psi

9. Pressure Test on Annulus:

- Close Dowell 2 x 2 valve in Annulus line and ensure annulus wing valve is closed. Pressure test the annulus surface line to 2500 psi against Dowell 2 x 2 valve. Observe for 10 minutes and bleed off line pressure.
- Open Dowell 2 x 2 valve and pressure lines to annulus wing valve to 3000 psi. Observe for 10 minutes and bleed off line pressure. Company man on location to confirm any discrepancies in maximum pressure limitations for testing.
- Open the isolation valve to the pressure relief valve and set PRV to 2000 psi. Pressure up line to confirm pop-off activation pressure. Reset Pop-off PRV. Re-test until activation pressure is exact.

10. Pressure up annulus to 1000 psi. Maintain annulus pressure at 1000 psi during DataFrac or as per DST Engineer recommendation. Record and report all volumes pumped and bled off during the operation.

11. Hold Safety meeting with all key personnel on location and assign the job responsibility

12. Open SLB Master Valve.

13. Pressure up annulus to 1000 psi. Maintain annulus pressure at 1000 psi during treatment. Record and report all volumes pumped and bleed off during the operation.

14. Perform the fracture treatment as per detailed fracture schedule in Appendix 2 or as redesigned on location

15. After Shut down, monitor the surface pressure till pressure reaches 500 psi below expected closure pressure. Close Schlumberger master valve to isolate frac pumps. Bleed off treating line.

- If the well screens out during the Main Frac Job, immediately S/D all pumps, bleed off both TBG & annulus pressures to zero. Do not close GATE valve. Then, proceed directly to the clean out operation using coiled tubing. (as per Schl. clean out program) till PBTB.

16. Flush Schlumberger surface treating line and frac pumps with water.

17. R/D Schlumberger frac surface lines & move equipments a way from well locations.

18. Bleed off the Annulus.

APPENDIX 1**SLB Equipment & Products Provision:**

- ⇒ 4 * 500 bbl frac tanks
- ⇒ Frac surface equipment, proppant, chemicals, etc :
- ⇒ Frac string down hole tools.

Water Analysis Requirements:

Perform analysis of water in each Frac Tank to verify gel can be mixed with the water and pre-mixing gel to verify crosslinking and delay time using water in each Frac Tanks and chemical on location.

- PH
- Temperature
- Carbonate and Bicarbonate
- Specific Gravity
- Iron concentration
- Pre-mixing, viscosity, pH, crosslinking and delay time.

AGIBA Provision:

- ⇒ 2000 bbls Fresh water
- ⇒ Enough lights for Coiled tubing night operation
- ⇒ Prepare flare pit to hold approximately 300 bbls of fluid. The pit should be down wind and approximately 150 ft away from the rig site.
- ⇒ Rig Safety fire fighting equip. should be available during the Frac job.

APPENDIX 2**FracCADE^{*}**
STIMULATION PROPOSAL

Operator : Agiba Petroelum Company
Well : Raml-17
Field : Raml Field
Formation : BIII

Well Location : Western Desert
City : Cairo
Country : Egypt

Prepared for : Agiba Petroleum Company **Service Point** : EEG
Proposal No. : 1 **Business Phone** : 0
Date Prepared : **FAX No.** : 02-3594526

Prepared by : Ahmed Samir Zaki
Phone : 0127-0001372
E-Mail Address : AZaki2@slb.com

* Mark of Schlumberger

Disclaimer Notice:

This information is presented in good faith, but no warranty is given by and Schlumberger assumes no liability for advice or recommendations made concerning results to be obtained from the use of any product or service. The results given are estimates based on calculations produced by a computer model including various assumptions on the well, reservoir and treatment. The results depend on input data provided by the Operator and estimates as to unknown data and can be no more accurate than the model, the assumptions and such input data. The information presented is Schlumberger's best estimate of the actual results that may be achieved and should be used for comparison purposes rather than absolute values. The quality of input data, and hence results, may be improved through the use of certain tests and procedures which Schlumberger can assist in selecting.

The Operator has superior knowledge of the well, the reservoir, the field and conditions affecting them. If the Operator is aware of any conditions whereby a neighboring well or wells might be affected by the treatment proposed herein it is the Operator's responsibility to notify the owner or owners of the well or wells accordingly.

Prices quoted are estimates only and are good for 30 days from the date of issue. Actual charges may vary depending upon time, equipment, and material ultimately required to perform these services.

Freedom from infringement of patents of Schlumberger or others is not to be inferred.

Section 1: Wellbore Configuration

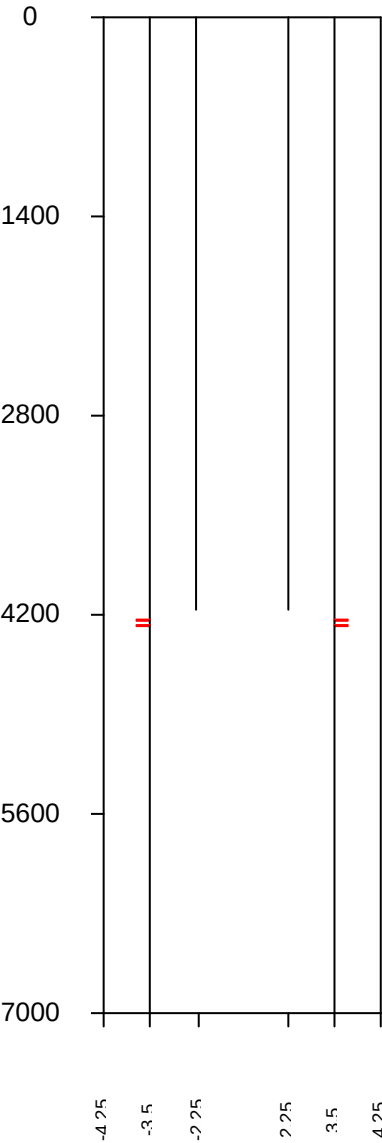
Bottom Hole Temperature.....150 degF
Deviated Hole.....NO
Treat Down.....TUBING

Well Type.....Vertical
Well Location.....OnShore

Tubing Data			
OD (in)	Weight (lb/ft)	ID (in)	Depth (ft)
4.500	15.5	3.830	4164.0

Casing Data			
OD (in)	Weight (lb/ft)	ID (in)	Depth (ft)
7.000	29.0	6.184	7000.0

Perforation Data						
Top MD (ft)	Top TVD (ft)	Bottom MD (ft)	Bottom TVD (ft)	Shot Density (shot/ft)	Number	Diameter (in)
4234.0	4234.0	4242.0	4242.0	5.00	40	0.6



Section 2: Propped Fracture Schedule

Pumping Schedule

The following is the Pumping Schedule to achieve a propped fracture half-length (X_f) of 315.9 ft with an average conductivity (K_{fw}) of 8469 md.ft.

Job Description						
Stage Name	Pump Rate (bbl/min)	Fluid Name	Stage Fluid Volume (gal)	Gel Conc. (lb/mgal)	Prop. Type and Mesh	Prop. Conc. (PPA)
Data Frac Schedule						
Pre Flush	30.0	F105 10%	3000	0.0		0.00
Spacer	30.0	Brine 4% KCl	2000	0.0		0.00
Calibration	30.0	ClearFRAC XT	7000	0.0		0.00
Flush	30.0	Brine 4% KCl	2300	0.0		0.00
Main Frac Schedule						
Pre Flush	30.0	F105 10%	3000	0.0		0.00
Spacer	30.0	Brine 4% KCl	2000	0.0		0.00
PAD	30.0	ClearFRAC XT	12000	0.0		0.00
1.0 PPA	30.0	ClearFRAC XT	3500	0.0	S580 20/40	1.00
2.0 PPA	30.0	ClearFRAC XT	3500	0.0	S580 20/40	2.00
3.0 PPA	30.0	ClearFRAC XT	3500	0.0	S580 20/40	3.00
4.0 PPA	30.0	ClearFRAC XT	3500	0.0	S580 20/40	4.00
5.0 PPA	30.0	ClearFRAC XT	4000	0.0	S580 20/40	5.00
6.0 PPA	30.0	ClearFRAC XT	4000	0.0	S580 20/40	6.00
Flush	30.0	Brine 4% KCl	2391	0.0		0.00

Please note that this pumping schedule is under-displaced by 5.0 bbl.

Fluid Totals		
6000 gal	of	Pre-Flush Fluid
8691 gal	of	Brine 4% KCl
41000 gal	of	ClearFRAC XT

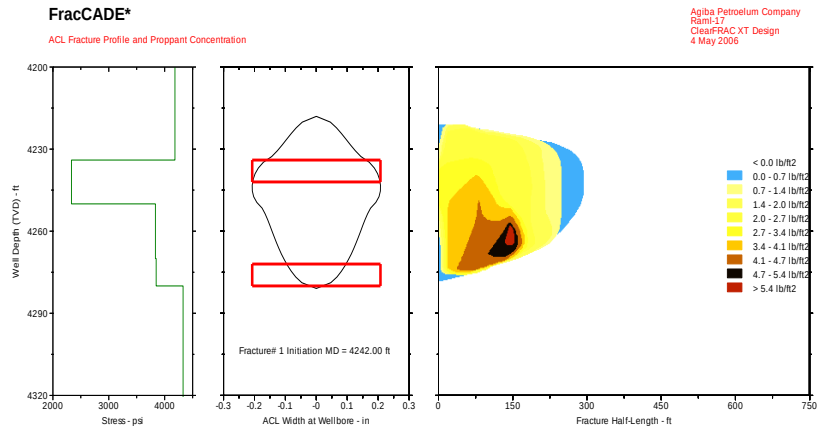
Proppant Totals		
79000 lb	of	S580 20/40

Pad Percentages	
% PAD Clean	35.3
% PAD Dirty	32.5

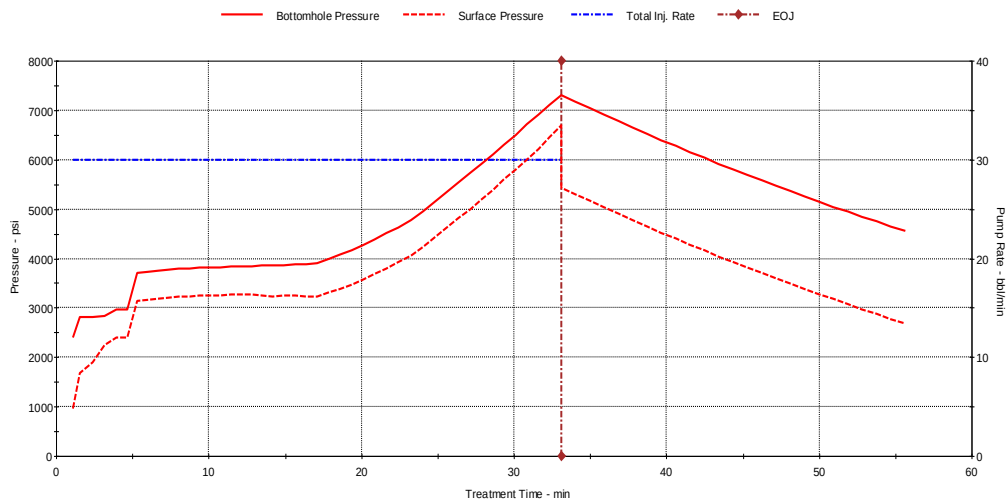
Job Execution									
Stage Name	Stage Fluid Volume (gal)	Cum. Fluid Volume (gal)	Stage Slurry Volume (bbl)	Cum. Slurry Volume (bbl)	Stage Prop (lb)	Cum. Prop. (lb)	Avg. Surface Pressure (psi)	Stage Time (min)	Cum. Time (min)
Pre Flush	3000	3000	71.4	71.4	0	0	954	2.4	2.4
Spacer	2000	5000	47.6	119.0	0	0	1321	1.6	4.0
PAD	12000	17000	285.7	404.8	0	0	2824	9.5	13.5
1.0 PPA	3500	20500	86.4	491.1	3500	3500	3255	2.9	16.4
2.0 PPA	3500	24000	89.4	580.6	7000	10500	3233	3.0	19.4
3.0 PPA	3500	27500	92.5	673.1	10500	21000	3392	3.1	22.4
4.0 PPA	3500	31000	95.5	768.6	14000	35000	3803	3.2	25.6
5.0 PPA	4000	35000	112.7	881.3	20000	55000	4524	3.8	29.4
6.0 PPA	4000	39000	116.2	997.5	24000	79000	5496	3.9	33.2
Flush	2391	41391	56.9	1054.4	0	79000	6337	1.9	35.1

Section 3: Propped Fracture Simulation Results

(1) ACL Fracture Profile and Proppant Concentration Plot



(2) Treating Plot



Section 4: Propped Fracture Simulation

The following are the results of the computer simulation of this Fracturing Proposal using a Pseudo 3-D Vertical model. Effective Conductivity and Effective Fcd are calculated based on perforated intervals with positive net heights.

Initial Fracture Top TVD	4234.0 ft
Initial Fracture Bottom TVD	4250.0 ft
Propped Fracture Half-Length	315.9 ft
EOJ Hyd Height at Well	62.9 ft
Average Propped Width	0.196 in
Average Gel Concentration	0.0 lb/mgal
Average Gel Fluid Retained Factor	0.98
Net Pressure	4979 psi
Efficiency	0.282
Effective Conductivity	17640 md.ft
Effective Fcd	8.6
Max Surface Pressure	6690 psi

Simulation Results by Fracture Segment							
From (ft)	To (ft)	Prop. Conc. at End of Pumping (PPA)	Propped Width (in)	Propped Height (ft)	Frac. Prop. Conc. (lb/ft ²)	Frac. Gel Conc. (lb/mgal)	Fracture Conductivity (md.ft)
0.0	79.0	7.4	0.246	42.9	2.21	0.0	11551
79.0	157.9	8.6	0.273	54.5	2.45	0.0	12774
157.9	236.9	7.7	0.242	47.1	2.17	0.0	9220
236.9	315.9	1.2	0.033	28.4	0.30	0.0	1678

Exposure Time Prediction by Stage						
Stage Name	Fluid Name	Pump Rate (bbl/min)	Fluid Volume (gal)	Perforation Injection Temp. (degF)	Exposure at BHST of 169 degF (min)	Exposure above Watch Temp. of 164 degF (min)
Pre Flush	Pre-Flush	30.0	3000	109	2.4	2.4
Spacer	Brine 4% KCl	30.0	2000	102	2.2	2.2
PAD	ClearFRAC XT	30.0	12000	88	6.8	6.8
1.0 PPA	ClearFRAC XT	30.0	3500	87	1.5	1.5
2.0 PPA	ClearFRAC XT	30.0	3500	86	0.0	0.0
3.0 PPA	ClearFRAC XT	30.0	3500	86	0.0	0.0
4.0 PPA	ClearFRAC XT	30.0	3500	86	0.0	0.0
5.0 PPA	ClearFRAC XT	30.0	4000	86	0.0	0.0
6.0 PPA	ClearFRAC XT	30.0	4000	86	0.0	0.0
Flush	Brine 4% KCl	30.0	2391			